

BORON FAMILY

Non-metal ← 5 B
Most abundant metal
Third most element in earth crust
(O > Si > Al > Fe)

13 Al
exists as Ga_2 molecules & thus low M.P (29.67° C)
-liquid up to 2000° C
Used in high T thermometer

Metal ← 31 Ga
49 In
81 Tl
113 Nh
Radioactive

PHYSICAL PROPERTIES

Atomic radii
B < Ga < Al < In < Tl

Ionisation enthalpy
B > Tl > Ga > Al > In

Electronegativity
B > Tl > In > Ga > Al

Ionic radii Decreases down the group

Density Increases Down the group

M.p decreases from B to Ga and then increases
B > Al > Tl > In > Ga

B.p Decreases down the group

CHEMICAL PROPERTIES

Reducing power Al > Ga > In > Tl
Stability order of O.S:
Tl⁺¹ > In⁺¹ > Ga⁺¹ > Al⁺¹ > B⁺¹ (**Inert pair effect**)
B⁺³ > Al⁺³ > Ga⁺³ > In⁺³ > Tl⁺³
Tl⁺³ = Strong oxidant, Ga⁺¹ = Strong reductant
Maximum covalency of B is 4 (absence of valence d orbitals)

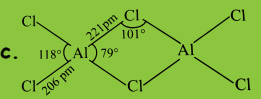
Halides -Lewis acids (BF₃ < BCl₃ < BBr₃)

Oxides	Hydroxides
Acidic Character Decreases Basic Character Increases	Acidic Character Decreases Basic Character Increases
B ₂ O ₃ Al ₂ O ₃ Ga ₂ O ₃ In ₂ O ₃ Tl ₂ O ₃	B(OH) ₃ Al(OH) ₃ Ga(OH) ₃ In(OH) ₃ Tl(OH) ₃

COMPOUNDS OF Al

AlCl₃

- Al₂O₃ + 3C + 3Cl₂ $\xrightarrow{\text{heat}}$ 2AlCl₃ + 3CO
- Aqueous solution is acidic due to the formation of HCl.
AlCl₃ + 3H₂O → Al(OH)₃ + 3HCl
- Anhyd. AlCl₃ is covalent & dimeric.



Alum

M₂SO₄ · M'(SO₄)₃ · 24H₂O
M - Monovalent metal M' - Trivalent metal
Examples:
Potash alum K₂SO₄ · Al₂(SO₄)₃ · 24H₂O
Chrome alum K₂SO₄ · Cr₂(SO₄)₃ · 24H₂O
Ferric alum (NH₄)₂ · SO₄ Fe₂(SO₄)₃ · 24H₂O
Each cation is surrounded by 6 H₂O

BORIC ACID

- Weak monobasic acid (Lewis acid)
H₃BO₃ + H₂O → [B(OH)₄]⁻ + H⁺
- Heating effect:
H₃BO₃ $\xrightarrow{273K}$ HBO₂ $\xrightarrow{433K}$ H₂B₄O₇ $\xrightarrow[\text{hot}]{\text{red}}$ B₂O₃
Metaboric acid Tetraboric acid Boron trioxide (Boric anhydride)
- Forms 6 H-bonds in aqueous solution.

DIBORANE

- Highly reactive: it catches fire
B₂H₆ + 3O₂ → B₂O₃ + 3H₂O, ΔH = -ve
- With water: B₂H₆(g) + 6H₂O(l) → 2B(OH)₃(aq) + 6H₂(g)
- Reaction with ammonia:
B₂H₆ + 2NH₃ $\xrightarrow[\text{Excess NH}_3]{\text{Low temp (-120°C)}}$ [BH₂(NH₃)₂]₂ $\xrightarrow[\text{Heat}]{\text{Borazine or borazole (inorganic benzene)}}$ 2B₃N₃H₆ + 12H₂

two 3c-2e banana bond
four 2c-2e Terminal bonds

BORAX / TINCAL

Na₂B₄O₇ · 10H₂O / Na₂[B₄O₇(OH)₄].8H₂O

- Aqueous solution of borax is alkaline in nature
Na₂B₄O₇ + 7H₂O → 2NaOH + 4H₃BO₃
- Borax bead test (detection of transition metal)
Na₂B₄O₇ · 10H₂O $\xrightarrow{\Delta}$ Na₂B₄O₇ $\xrightarrow{740^\circ\text{C}}$ 2NaBO₂ + B₂O₃
Sodium Metaborate Boric anhydride
Glassy mass (Borax bead)

T.M	Colour
Cu/Co	Blue

Uses: As flux
Water softening agent
As antiseptic
In making enamel
Manufacture of glass

• There are 5 B-O-B bridge bonds

CARBON FAMILY

- 1) Oxidation State: +2, +4, -2
Stability of +4: C > Si > Ge > Sn > Pb
Stability of +2: Pb > Sn > Ge > Si > C
- 2) Oxides
- | | |
|---------------------------|-----------------------------|
| CO Neutral | CO ₂ Acidic |
| SiO Neutral | SiO ₂ Acidic |
| GeO Acidic | GeO ₂ Acidic |
| SnO Amphoteric | SnO ₂ Amphoteric |
| PbO Amphoteric (Litharge) | PbO ₂ Amphoteric |
- Halide**
- Thermal stability order:
CCl₄ < SiCl₄ < GeCl₄ < SnCl₄ < PbCl₄
CCl₄ > SiCl₄ > GeCl₄ > SnCl₄ > PbCl₄
 - CCl₄ can't be hydrolysed due to absence of valence d orbitals
 - PbI₄ & PbBr₄ does not exist due to strong oxidising nature of Pb⁺⁴

COMPOUNDS OF CARBON

CO

- HCOOH $\xrightarrow{\text{conc. H}_2\text{SO}_4}$ CO + H₂O (100% pure)
- Coal gasification
C + H₂O $\xrightarrow[\text{Syn gas / Water gas}]{\text{(steam)}}$ CO + H₂
- Producer gas (CO + N₂)
- With Hb it forms 300 times stable Carboxy Hb (w.r. to oxyHb)

CO₂

- Solid CO₂ - Dry ice (Refrigerant)
- CO₂ in water gives carbonic acid (maintain pH 7.26-7.42)
- NaOH + CO₂ → NaHCO₃
- Used in soft drinks
- Used as fire extinguisher

P-BLOCK ELEMENTS

COMPOUNDS OF SILICON

SILICA

- SiO₂ - sand, quartz (piezoelectric)
- Insoluble in H₂O & inert at room temp.
- Reacts with HF (etching of glass)
HF + SiO₂ → SiF₄
SiF₂ + 2HF → H₄SiF₄
- 3D Network Covalent Solid

SILICONES

- general formula (R₂SiO)_n
- R₂SiCl₂ + H₂O → R₂Si(OH)₂
(Linear Chain Silicones)
- RSiCl₃ $\xrightarrow{\text{H}_2\text{O}}$ RSi(OH)₃
(Cross linked silicones)
- R₃SiCl → stopping agent / Dimer

SILICATES

- Metal derivatives of silicic acid H₄SiO₄
- Basic unit is (SiO₄)⁴⁻ Tetrahedral

ZEOLITES

- Sodium Aluminium silicates (Na₂Al₂Si₂O₈ · XH₂O)
- (i) Used for Purification of H₂O to remove hardness of water
- (ii) ZSM 5 is a shape selective catalyst to convert alcohol to gasoline

ALLOTROPES OF CARBON

DIAMOND

- C-C bond length is 154 pm
- C is sp³ hybridised
- Good thermal conductor
- ΔH of formation is 1.9KJ/mol
- Used as abrasive for sharpening of tools

GRAPHITE

- C-C bond length is 141.5pm
- Thermodynamically most stable due to Δ_rH = 0
- C is sp² hybridised
- Good electric conductor
- Hexagonal ring layers which are 340pm apart
- Used as dry lubricant in machines

FULLERENES

- C₆₀-C₇₀ (Bucky balls)
- C is sp² hybridised
- C-C → 143.5pm
C=C → 138.3pm
- C₆₀ has 12 pentagons and 20 hexagons